

# Individual Assignment 5

Alyssa Hagele

## 1. Set Up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

### Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

## Class code

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),

```

```

        med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>

```

```

# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
  site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
  fun = "median",
  size = 4,
  color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title

```

```

labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

# base layer
salinity <- ggplot(data = aq_ins_clean,
                  # x-axis: site
                  # y-axis: mean salinity
                  mapping = aes(x = site_full,
                               y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
            color = "red",
            size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

```

## 2. Problems

### 1. Visualizing differences across sites in major taxon abundance

#### a. Plan your figure

- x-axis: site\_full
- y-axis: log\_ave\_lit
- geometry to represent average abundance/L: geom\_jitter()
- geometry to represent medians: stat\_summary()

## b. Wrangle your data

```
#creating new object frm aq_ins_clean
ostracods <- aq_ins_clean |>
  #filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#display 10 random observations
ostracods |>
  slice_sample(n = 10)
```

# A tibble: 10 x 24

|    | date_site<br><chr> | date<br><dtm>       | site<br><chr> | taxon<br><chr> | ave_lit<br><dbl> | med_ph<br><dbl> | med_sal<br><dbl> | med_do<br><dbl> |
|----|--------------------|---------------------|---------------|----------------|------------------|-----------------|------------------|-----------------|
| 1  | 2021-08-25_NPB     | 2021-08-25 00:00:00 | NPB           | ostr~          | 581.             | 7.52            | 4.57             | 4.5             |
| 2  | 2023-09-15_NMC     | 2023-09-15 00:00:00 | NMC           | ostr~          | 0.4              | NA              | NA               | 14.2            |
| 3  | 2022-02-21_NPB     | 2022-02-21 00:00:00 | NPB           | ostr~          | 1.33             | 7.39            | 34.1             | 0.34            |
| 4  | 2023-02-21_NEC     | 2023-02-21 00:00:00 | NEC           | ostr~          | 0                | 8.56            | NA               | 8.8             |
| 5  | 2022-08-10_NPB     | 2022-08-10 00:00:00 | NPB           | ostr~          | 2.67             | 8.28            | 8.84             | 9.81            |
| 6  | 2023-07-17_NPB     | 2023-07-17 00:00:00 | NPB           | ostr~          | 4.53             | NA              | NA               | NA              |
| 7  | 2023-02-28_NPB1    | 2023-02-28 00:00:00 | NPB1          | ostr~          | 0                | NA              | NA               | NA              |
| 8  | 2021-04-15_NPB     | 2021-04-15 00:00:00 | NPB           | ostr~          | 185.             | 8.09            | 3.57             | 8.24            |
| 9  | 2023-05-16_NVBR    | 2023-05-16 00:00:00 | NVBR          | ostr~          | 0.133            | 9.29            | NA               | 9.83            |
| 10 | 2023-05-05_NPB     | 2023-05-05 00:00:00 | NPB           | ostr~          | 24.7             | NA              | NA               | 12.1            |

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

## c. Code your figure

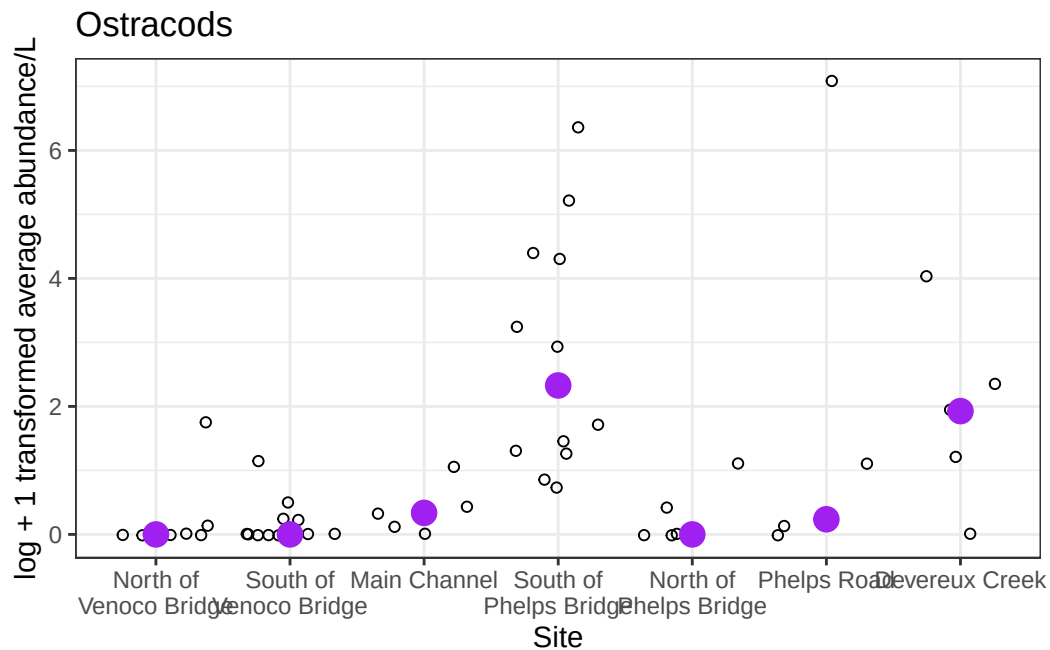
```
# base layer
ostracods_plot <- ggplot(data = ostracods,
  # x-axis: site
  # y-axis: log + 1 transformed average abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
```

```

# first layer: points
geom_jitter(shape = 21) +
# second layer: median points
# changing size, color, shape
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "purple") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# custom theme
theme_bw()

ostracods_plot

```



#### d. Create a multi panel figure

```
#using patchwork to create multipanel figure
# *used chatgpt for help*
salinity /
  (copepods | ostracods_plot)
```

